

Visual Misinformation Detection: Out-of-Context and Manipulation Cues via CLIP and ResNet-18

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Abstract

I present a two-stage visual pipeline for detecting image-based misinformation. Stage 1 identifies out-of-context images by measuring the cosine similarity between CLIP image and text embeddings: a pristine news image should be semantically close to its caption, while a recycled image paired with a fabricated caption is not. Evaluated on Flickr30k, Stage 1 achieves 98.9% accuracy and 0.999 AUC-ROC with no task-specific training. Stage 2 detects image manipulation by fine-tuning ResNet-18 on the CASIA v2 forgery dataset, reaching 77.6% accuracy and 0.847 AUC-ROC. Grad-CAM visualizations show the model attends to spliced regions rather than global texture. Together the two stages address complementary forms of visual deception and form the CV component of a broader multi-modal fake-news detection system.

1. Introduction

Images in online news can mislead in two distinct ways. First, an authentic photograph may be taken out of context, recycled from an unrelated event and paired with a fabricated caption. Second, the image itself may be digitally altered: regions spliced, objects removed, or faces swapped. Both forms are common in misinformation campaigns, yet they require different detection strategies.

Fact-checking a recycled image requires reasoning about image-text *semantic coherence*, a task for vision-language models. Detecting pixel-level forgery, by contrast, requires attending to local artifacts such as inconsistent boundaries, noise patterns, and compression traces; a task better suited to convolutional features. I therefore design a two-stage pipeline: Stage 1 uses CLIP cosine similarity for out-of-context detection, and Stage 2 fine-tunes ResNet-18 on CASIA v2 for manipulation detection. This report describes both stages, their datasets, training details, and results.

2. Related Work

Vision-language grounding. CLIP [1] learns aligned image and text representations via contrastive pre-training on 400M pairs, enabling zero-shot retrieval and classification. Luo et al. [2] build NewsCLIPpings by pairing news images with semantically plausible but incorrect captions and show that CLIP features are strong out-of-context cues. I follow a similar approach but use Flickr30k for reproducibility and evaluate purely on cosine similarity without additional fine-tuning.

Image forgery detection. Early methods [6] rely on copy-move detection and JPEG artifact analysis. More recent work fine-tunes CNNs on CASIA v2 [3], which remains a standard benchmark. I adopt this approach with ResNet-18 as a compact baseline and use Grad-CAM [5] to validate that the model attends to the manipulated regions.

3. Datasets

Flickr30k (Stage 1). I use the `clip-benchmark/wds_flickr30k` test split from HuggingFace (1,000 images, each with one caption). I construct 2,000 image-text pairs: 1,000 *pristine* (image + its own caption) and 1,000 *out-of-context* (image + a caption from a different image, shuffled to avoid self-pairing). This mirrors the NewsCLIPpings construction methodology.

CASIA v2 (Stage 2). The CASIA image manipulation dataset contains 7,491 authentic and 5,123 tampered images across diverse categories including copy-move and splicing. I split 80/20 stratified by label, giving 10,093 training images and 2,523 test images.

4. Method

4.1. Stage 1 | Out-of-Context Detection

I load `openai/clip-vit-base-patch32` and extract L2-normalized image and text embeddings for all 2,000 pairs without any fine-tuning. The credibility score is the dot product of the (normalized) embeddings, i.e. the cosine similarity. A high score indicates that the image matches its caption; a low score suggests the pair is out-

of-context. I threshold at the median similarity to produce binary predictions. The AUC-ROC is computed over the continuous similarity scores to measure ranking quality independently of the threshold.

4.2. Stage 2 | Manipulation Detection

I fine-tune ResNet-18 [4] (ImageNet pretrained) by replacing its final layer with a 2-class linear head. Training uses AdamW ($\text{lr} = 10^{-4}$, weight decay 10^{-4}), CosineAnnealing LR schedule over 10 epochs, and cross-entropy loss. Input images are resized to 224×224 ; training augmentations include random horizontal flip, ± 10 rotation, and color jitter (brightness/contrast ± 0.2). I keep the best checkpoint by validation AUC-ROC.

Grad-CAM. After training I run Grad-CAM on the final convolutional layer to produce saliency maps, which I visually inspect to verify that the model focuses on spliced boundaries.

5. Experiments and Results

5.1. Stage 1 Results

Metric	Out-of-ctx	Pristine
Precision	0.99	0.99
Recall	0.99	0.99
F1	0.99	0.99
Accuracy		0.989
AUC-ROC		0.999
Avg. Precision		0.999

Table 1. Stage 1 (CLIP zero-shot, 2,000 pairs).

The CLIP similarity scores are already very well separated: the score distribution (Figure 1) shows a clear bimodal structure with almost no overlap between pristine and out-of-context pairs. This confirms that a simple cosine threshold is sufficient without any task-specific fine-tuning. The high AUC (0.999) indicates that the separation holds across all possible thresholds.

5.2. Stage 2 Results

Class	Prec.	Rec.	F1
Authentic	0.85	0.76	0.80
Tampered	0.69	0.80	0.74
Accuracy			0.776
F1-macro			0.773
AUC-ROC			0.847

Table 2. Stage 2 ResNet-18 on CASIA v2 test set (2,523 images).

Figure 2 shows the ROC curve and confusion matrix. The model achieves 0.847 AUC-ROC; meaningfully above

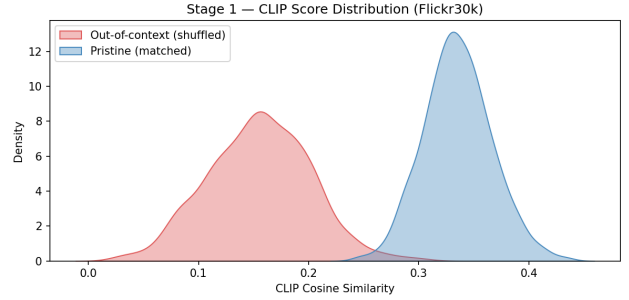


Figure 1. CLIP cosine similarity distributions for pristine (blue) and out-of-context (red) pairs. The distributions are almost non-overlapping.

random but with room for improvement. Recall on tampered images (0.80) is higher than precision (0.69), reflecting a bias toward predicting tampering. This is preferable in a security context (fewer missed forgeries), though it produces false alarms on authentic images. The 24% miss-rate on authentic samples is the main failure mode.

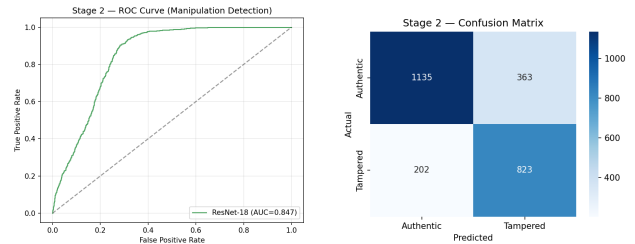


Figure 2. Left: ROC curve for Stage 2 (AUC=0.847). Right: confusion matrix on the CASIA v2 test set.

Grad-CAM analysis. Figure 3 shows representative saliency maps. For correctly classified tampered images, the model highlights the spliced boundary region rather than the image background, suggesting it learns local artifact cues. For the two misclassified authentic samples the activations are diffuse, consistent with genuine images that happen to have noisy textures.

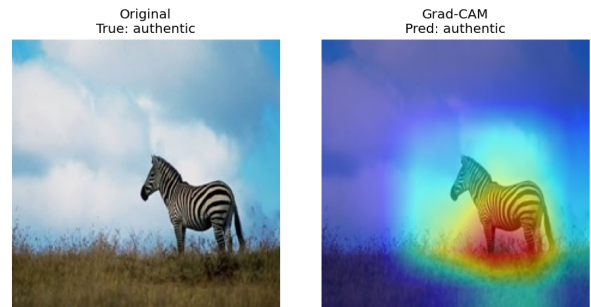


Figure 3. Grad-CAM saliency (authentic, correctly classified). Activations are spread rather than concentrated on a specific region.

6. Conclusion

Stage 1 (CLIP zero-shot) essentially solves out-of-context detection on the Flickr30k benchmark (AUC=0.999), confirming that off-the-shelf vision-language alignment is a near-perfect signal for semantic mismatch between image and caption. Stage 2 (ResNet-18 on CASIA v2, AUC=0.847) demonstrates that transfer learning from ImageNet is a viable starting point for pixel-level forgery detection, though the gap from Stage 1 highlights that manipulation detection remains the harder subproblem.

Limitations: Stage 1 is evaluated on clean image-caption pairs from Flickr30k; real misinformation likely involves more subtle semantic mismatches. Stage 2 is trained and tested on CASIA v2 only and may not generalize to GAN-generated or AI-upscaled content. Future work could add a transformer-based forgery detector (e.g. SplicedFormer) and integrate both stages with the LLM text credibility score into a joint multimodal verdict.

References

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